

What the world's worst outbreak can teach its neighbours

A COVID-19 data story

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Johns Hopkins CSSE confirmed cases · 22 Jan 2020 - 9 Mar 2023

The US recorded **103 million** cases. The question that matters for its neighbours:
could they have seen their own waves coming?

The approach

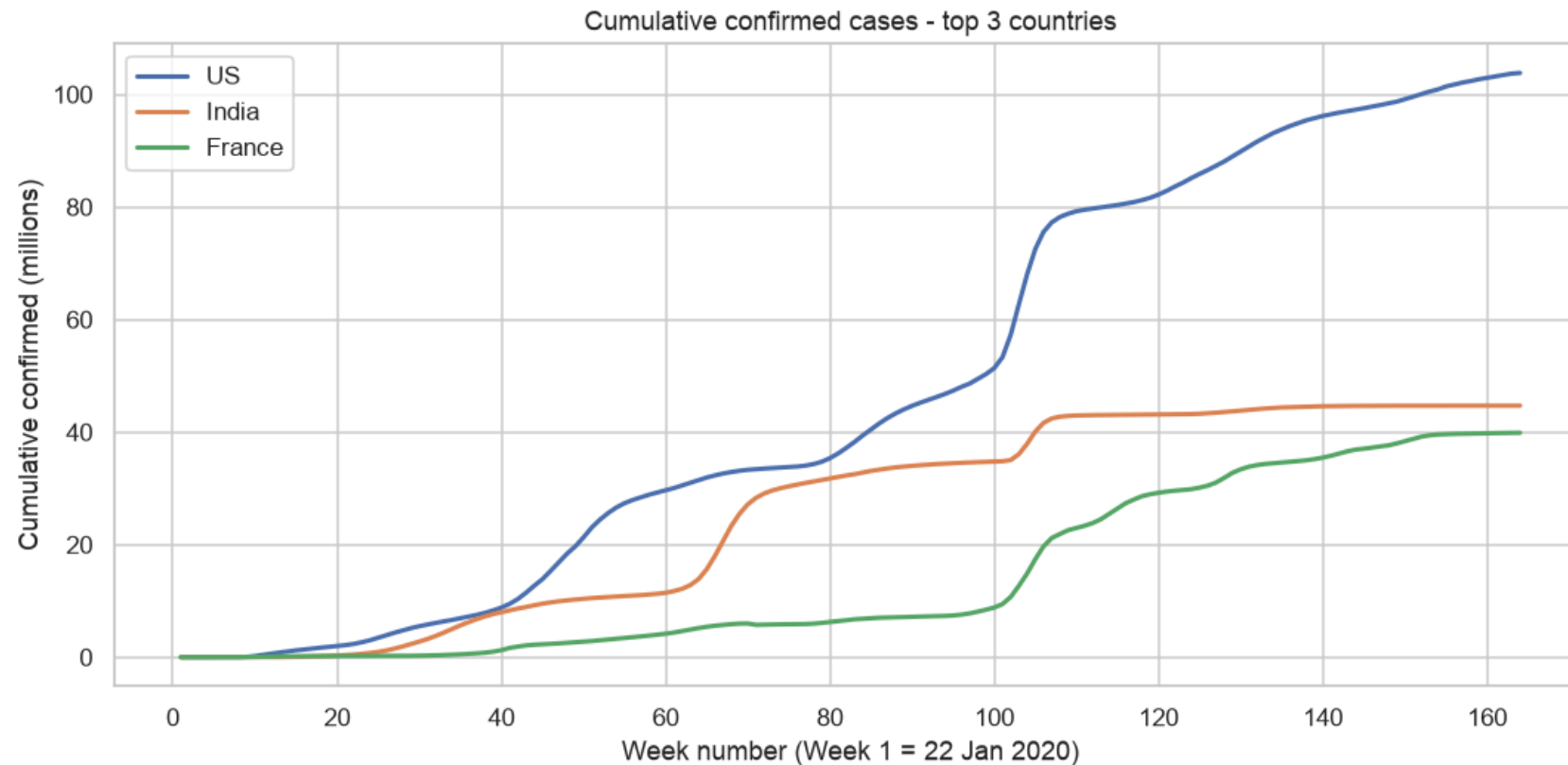
Every big-data project runs **prepare → analyse → decide**.

One analytical chain, each step feeding the next:

**top-3 countries → linear regression → pick the most volatile → K-Means waves → graph to neighbours
→ recommendation**

Tools: **Apache Spark MLlib** (regression + K-Means) · **networkx** (graph) · 164 weeks of data.

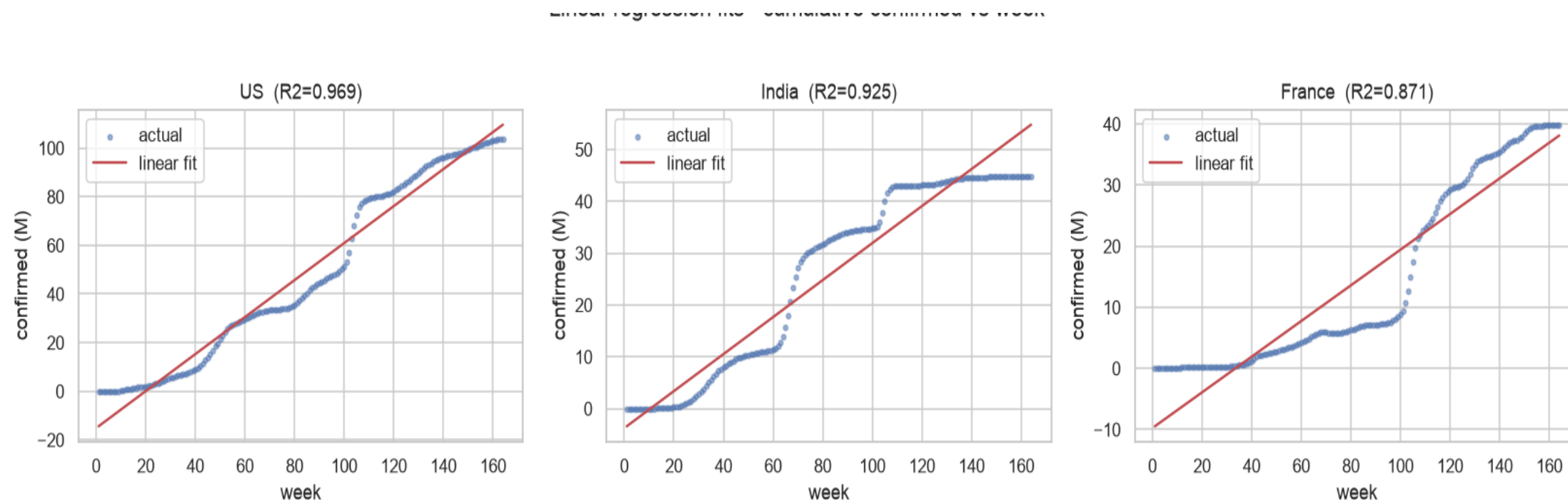
The three worst-hit countries



Top 3 by total confirmed: **US 103.8M · India 44.7M · France 39.9M.**

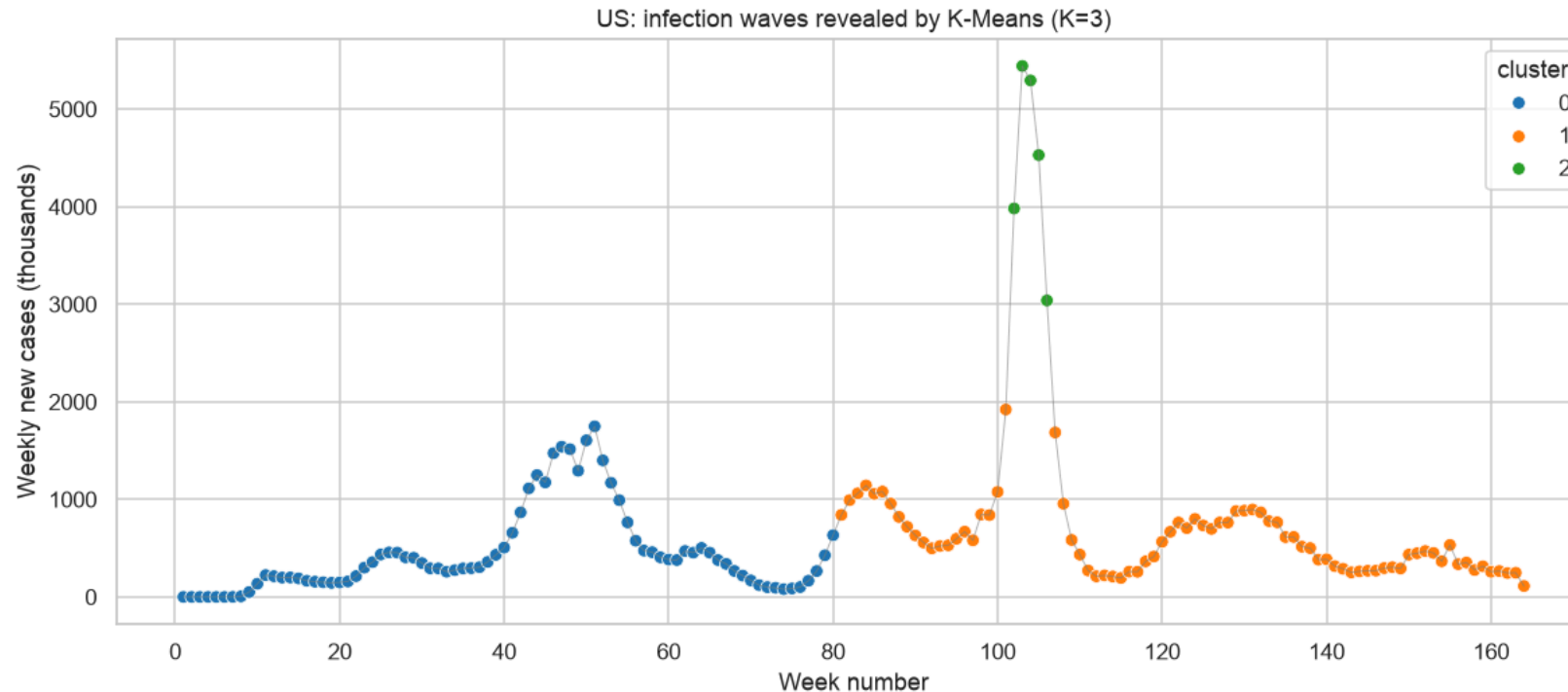
All three rise - but at very different rates and shapes. Which one is the *most volatile*?

Predictive modelling: a line is not enough



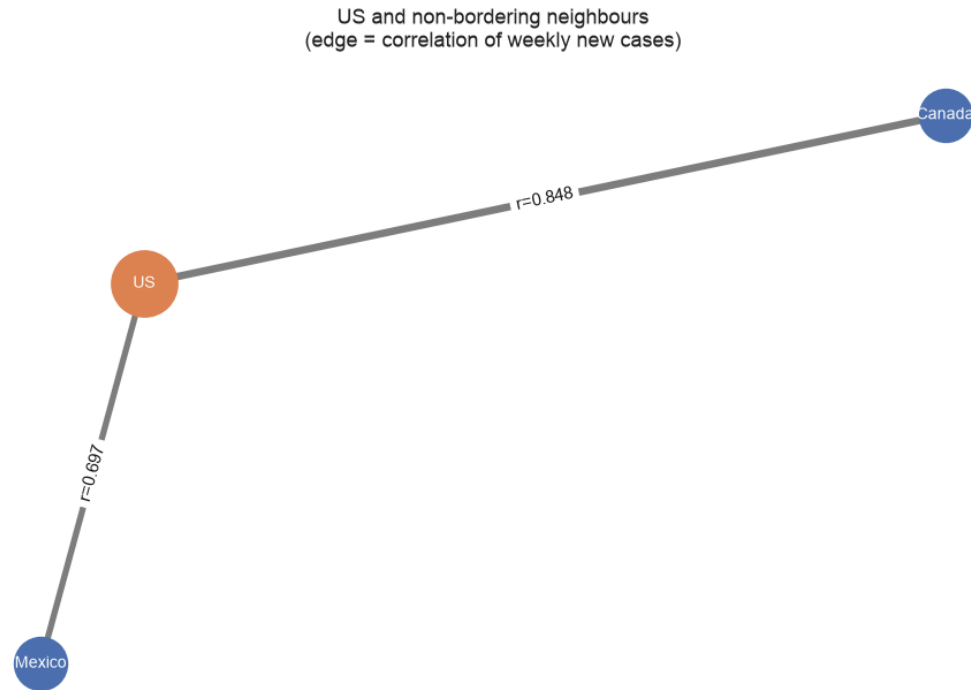
- Linear regression of cumulative cases on week number, per country.
- **US is the most volatile** - by the variance of its *weekly new cases* ($6.4e11$ vs India $2.5e11$, France $1.6e11$), not merely its size. Slope $\sim 760k$ cases/week, $R^2 = 0.97$.
- Key point: even $R^2 = 0.97$ **hides** the story - a straight line cannot show *when* the surges hit.

Clustering reveals the waves



- K-Means on [week, weekly new cases] , best $K = 3$ (silhouette 0.705).
- It isolates a **mega-surge of ~4.46M new cases/week around weeks 102-106** (the Omicron wave, Jan 2022) as its own cluster.
- This **proves** growth was not steady: it came in waves (up - down - up).

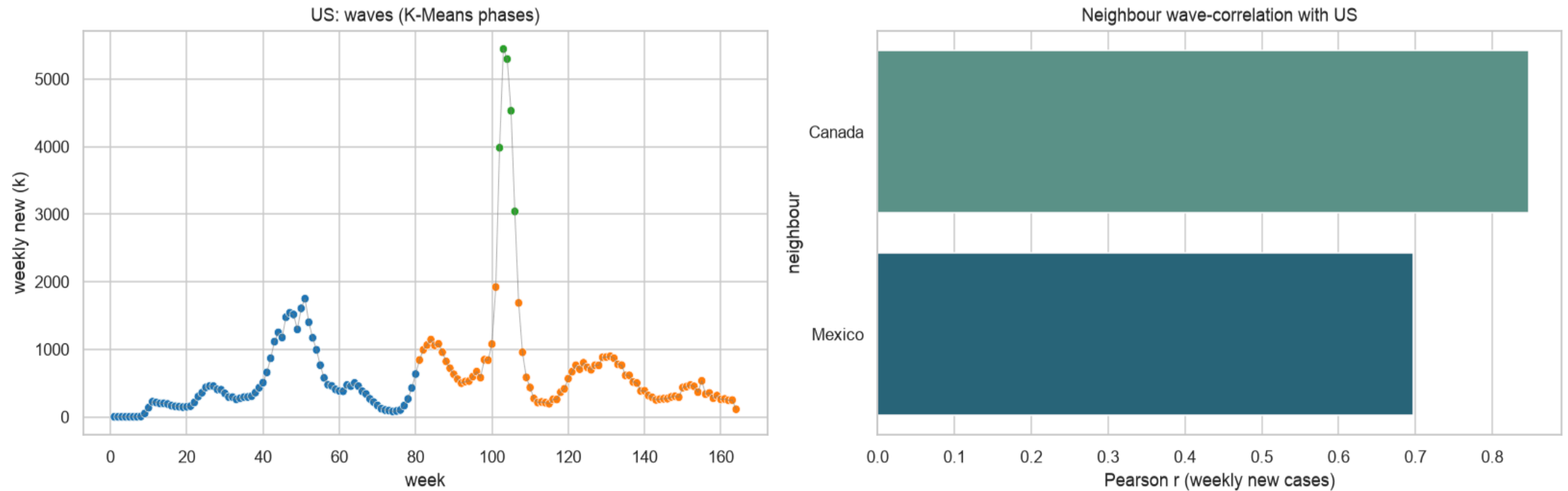
Graph analytics: who moves with the US?



- US linked to non-bordering neighbours **Canada** and **Mexico** (they do not border each other).
- Edge = correlation of weekly new cases: **Canada** $r = 0.85$ (strong), **Mexico** $r = 0.70$.
- Canada's waves track the US most closely → the strongest early-warning candidate.

The whole story in one frame

From raw cases to a neighbour early warning recommendation



Left: the US waves by phase. Right: how strongly each neighbour correlates.

The line from raw data → phases → a neighbour recommendation.

Recommendations to the neighbours

- **Canada ($r = 0.85$):** treat the US trajectory as a leading indicator; when the US enters a surge phase, pre-position testing and hospital capacity **1-2 weeks ahead**.
- **Mexico ($r = 0.70$):** moderate coupling - watch US surges, but weight local signals more heavily.
- **General:** plan capacity for the isolated **Omicron-style mega-surge cluster**, not the steady baseline - that single cluster carried $\sim 7x$ the overall weekly average.

Limitations & close

- **Data:** counts are cumulative and reporting-dependent; the JHU series **stopped 10 Mar 2023**.
- **Method:** `week` is a clustering input, so the phases are partly temporal by construction; "neighbour" is defined by geography, not true population mobility.
- **Next steps:** model weekly new cases directly, add mobility/vaccination data, and test lead-lag (cross-correlation) to quantify the warning window.

Close: data turned a wall of numbers into a concrete, actionable warning for neighbours.